

# ClamAV: A Free and Open Source Antivirus Tool



ClamAV is a free and open source toolkit to detect malware. It also performs Web/email scanning and provides gateway security for Linux distributions, in particular. Here's a simple guide on how to install and use this tool.

**W**e know that Linux is more secure than Windows, because of which many people think that we don't require antivirus software in Linux. But the fact is that viruses and malware do infiltrate Linux systems too. Though, it is true that the risk is lower compared to Windows. Personally, I haven't found any noxious intruders in my Linux box yet, but we can't say that it will never happen. So it's better to take some precautions to avoid any kind of attack.

## When should ClamAV be used?

- When you have very sensitive data and hence don't want to take any risks, ClamAV will provide an additional level of security.
- Use it when you want to do a system scan without booting into the system, so that viruses and malware do not get activated during the scan.
- When scanning external mails for any malware, since ClamAV is more helpful as a gateway scanner.

```
neulth@neulth-VirtualBox:~$ sudo apt-get update
[sudo] password for neulth:
Get:1 http://security.ubuntu.com/ubuntu xenial-security InRelease [102 kB]
Get:2 http://ft.archive.ubuntu.com/ubuntu xenial InRelease
Get:3 http://ft.archive.ubuntu.com/ubuntu xenial-updates InRelease [102 kB]
Get:4 http://ft.archive.ubuntu.com/ubuntu xenial-backports InRelease [102 kB]
Get:5 http://security.ubuntu.com/ubuntu xenial-security/main i386 DEP-11 Metadata
  # [48.1 kB]
Get:6 http://security.ubuntu.com/ubuntu xenial-security/main DEP-11 i386 i386
  [43.1 kB]
Get:7 http://ft.archive.ubuntu.com/ubuntu xenial-updates/main i386 DEP-11 Metadata
  [308 kB]
Get:8 http://security.ubuntu.com/ubuntu xenial-security/universe i386 DEP-11 Met
  adata [32.5 kB]
Get:9 http://security.ubuntu.com/ubuntu xenial-security/universe DEP-11 i386 i386
  [21.8 kB]
```

Figure 1: Updating the repository

```
neulth@neulth-VirtualBox:~$ sudo apt-get install clamav clamav-daemon
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  clamav-base clamav-freshclam clamscan libclamav7 liblinv3.6v5 libtmpack8
Suggested packages:
  clamav-docs clamav-libclamav-r7
The following NEW packages will be installed:
  clamav clamav-base clamav-daemon clamav-freshclam clamscan libclamav7
liblinv3.6v5 libtmpack8
0 upgraded, 8 newly installed, 0 to remove and 426 not upgraded.
Need to get 18.4 MB of archives.
After this operation, 42.5 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Figure 2: Installing ClamAV

```
neulth@neulth-VirtualBox:~$ sudo freshclam
ERROR: /var/log/clamav/freshclam.log is locked by another process
ERROR: Problem with internal logger (UpdateLogfile = /var/log/clamav/freshclam
  log).
```

Figure 3: Error while updating virus database

```
neulth@neulth-VirtualBox:~$ ps -aux | grep freshclam
clamav 1504 0.0 0.0 74800 14960 ? Ss 04:55 0:07 /usr/bin/fresh
  clam -d -foreground -r
neulth@neulth-VirtualBox:~$ ps -aux | grep freshclam
clamav 1547 0.0 0.0 5388 804 pts/1 S+ 05:11 0:00 grep --color
  -e freshclam
neulth@neulth-VirtualBox:~$ kill 1547
bash: kill: (1547) - Operation not permitted
neulth@neulth-VirtualBox:~$ sudo kill 1547
neulth@neulth-VirtualBox:~$ sudo freshclam
[info] update process started at Mon Feb 6 05:15:26 2017
[info] Downloading main.cvd [ 0%]
```

Figure 4: Updating virus database

```
neulth@neulth-VirtualBox:~$ sudo freshclam
[sudo] password for neulth:
ClamAV update process started at Mon Feb 6 14:48:00 2017
Downloading main.cvd [100%]
main.cvd updated (version: 57, sigs: 4218790, f-level: 68, builder: antihacker)
Downloading daily.cvd [100%]
daily.cvd updated (version: 23812, sigs: 1589882, f-level: 63, builder: neo)
Downloading bytecode.cvd [100%]
bytecode.cvd updated (version: 290, sigs: 55, f-level: 63, builder: neo)
```

Figure 5: Virus database update completed

## ClamAV installation

As ClamAV is open source, many third parties have developed different versions of it for different operating systems.

Let's look at how we can install it in Ubuntu. First, update the repository packets lists as follows (this is optional):

```
sudo apt-get update
```

Issue the command given below to install ClamAV. It will install *clamav-freshclam* also.

```
sudo apt-get install clamav
clamav-daemon
```

Now ClamAV is installed in our system. The next step is to update the virus definition database. This process is similar to normal updates done when instructing any antivirus software to fetch the latest virus related information. Once we run the command given below, two files -- *main.cvd* and *daily.cvd* -- will be downloaded and the virus database will be updated.

```
sudo freshclam
```